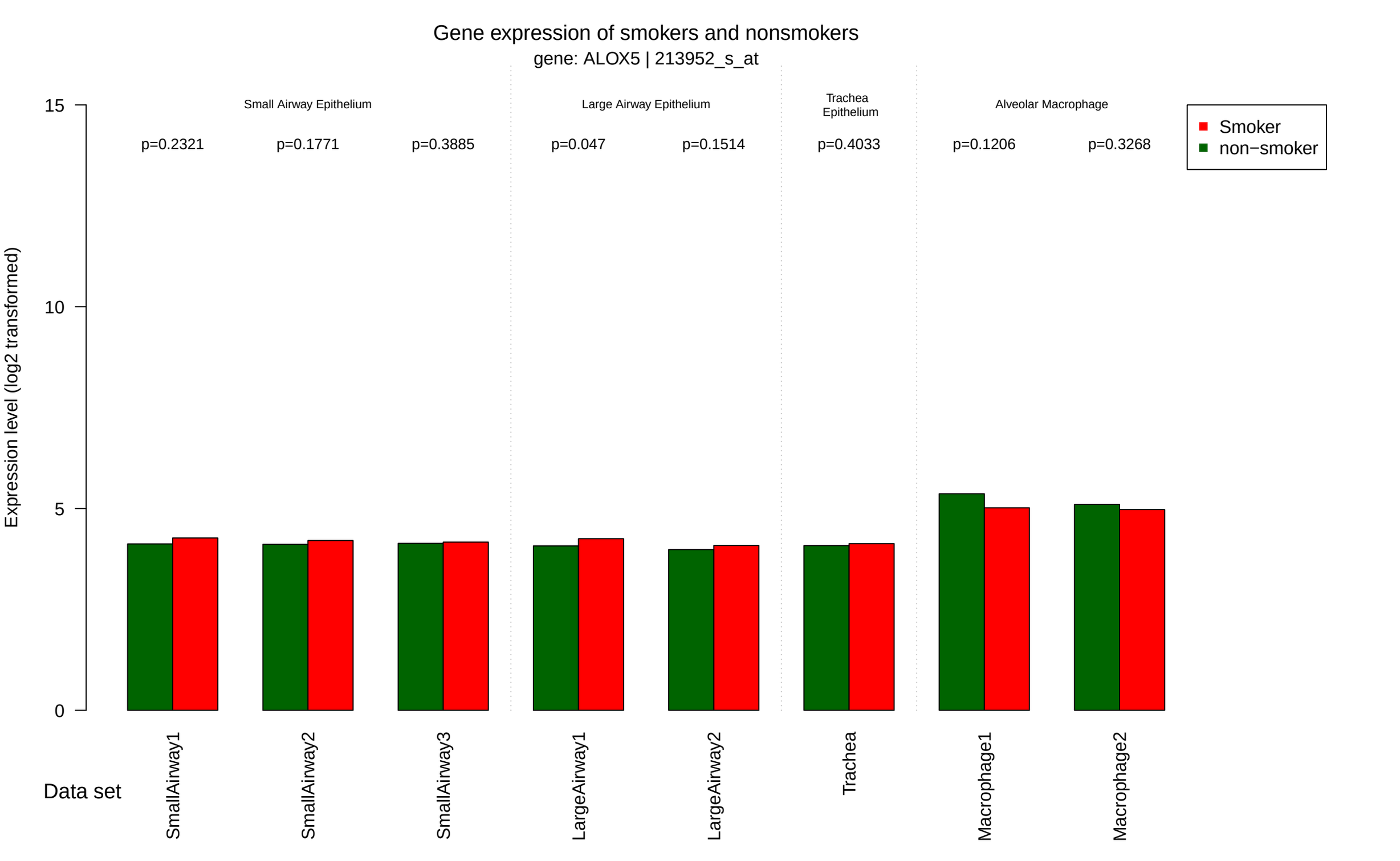
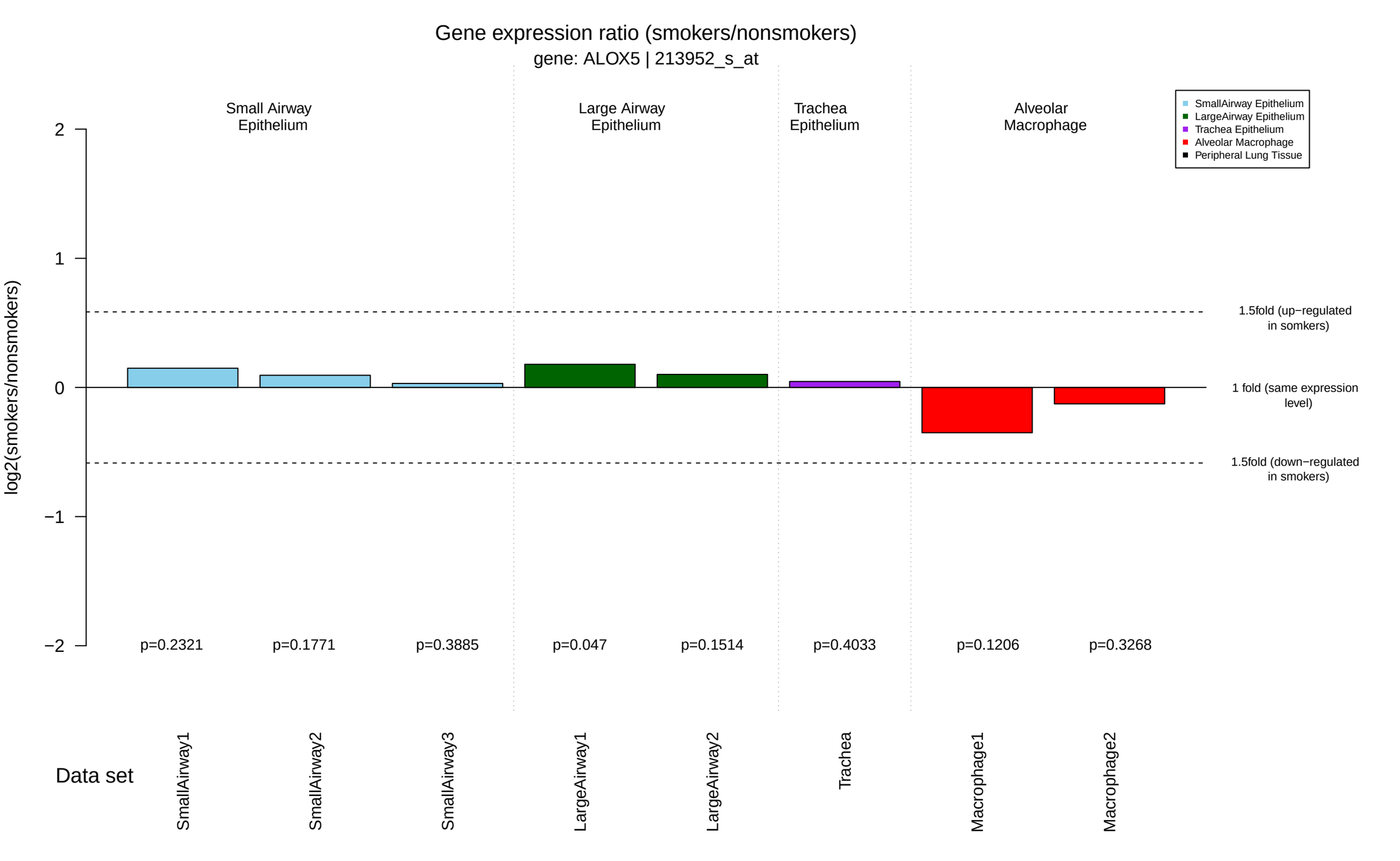
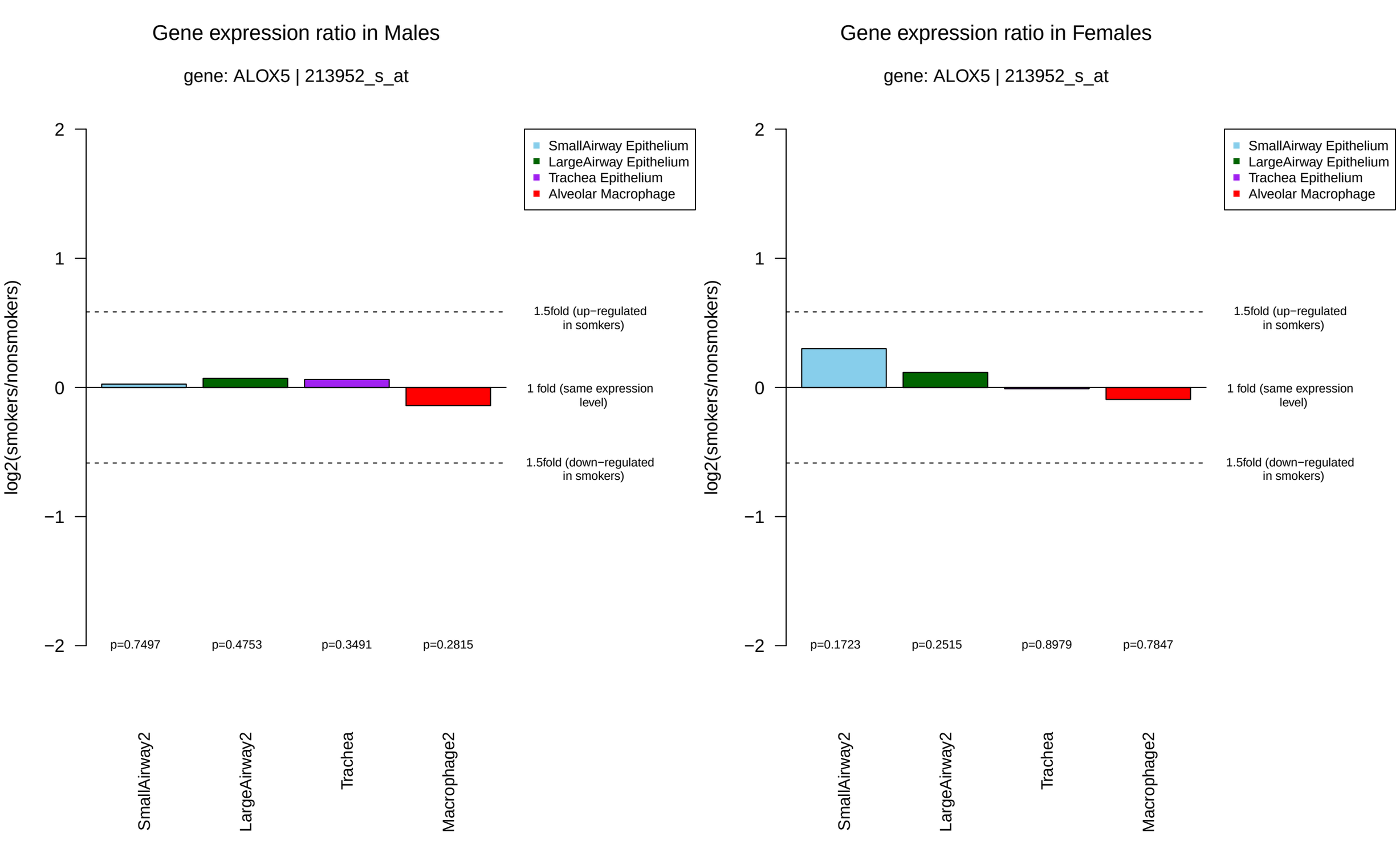
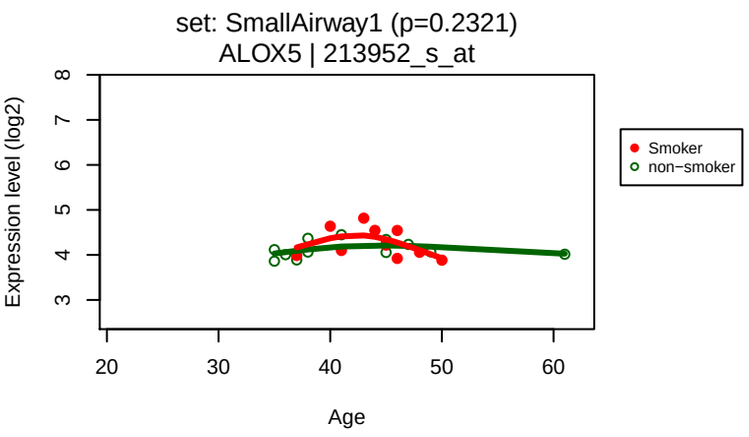
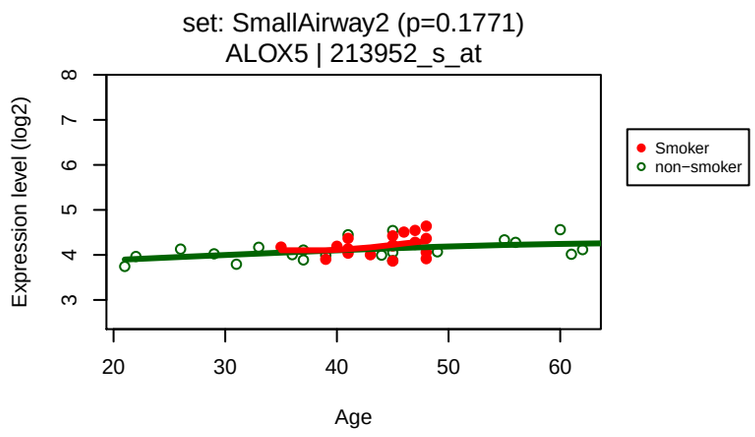
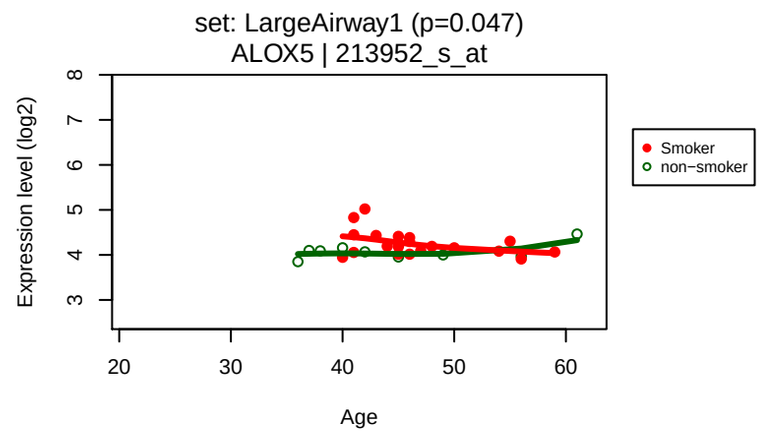

Gene expression of smokers and nonsmokers
gene: ALOX5 | 213952_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2321
p=0.1771
p=0.3885
p=0.047
p=0.1514
p=0.4033
p=0.1206
p=0.3268
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

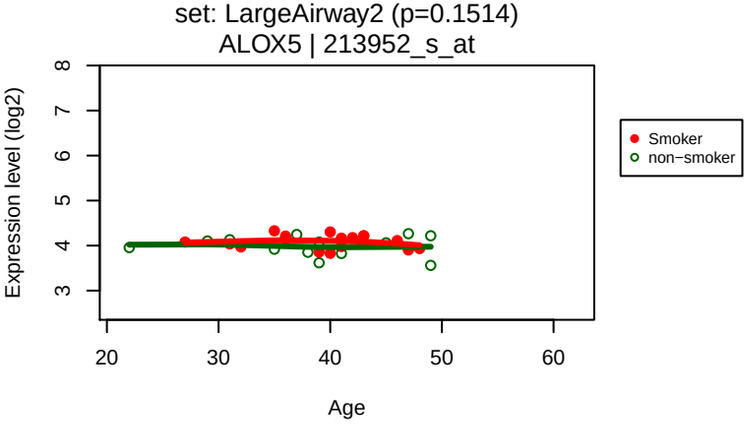

Gene expression ratio (smokers/nonsmokers)
gene: ALOX5 | 213952_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2321
p=0.1771
p=0.3885
p=0.047
p=0.1514
p=0.4033
p=0.1206
p=0.3268
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

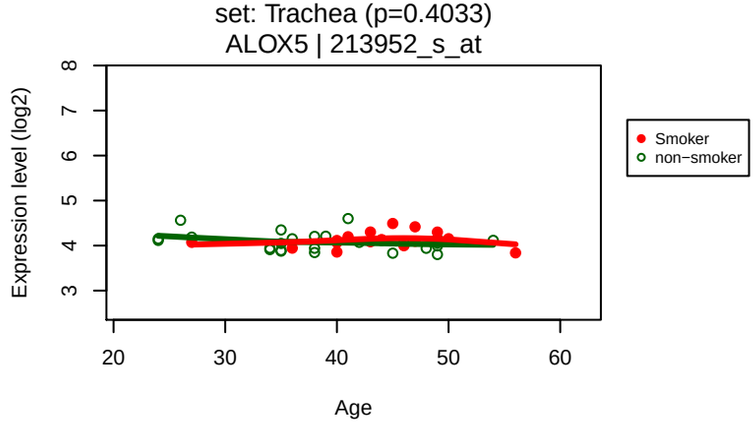

Gene expression ratio in Males
gene: ALOX5 | 213952_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7497
p=0.4753
p=0.3491
p=0.2815
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOX5 | 213952_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1723
p=0.2515
p=0.8979
p=0.7847
SmallAirway2
LargeAirway2
Trachea
Macrophage2

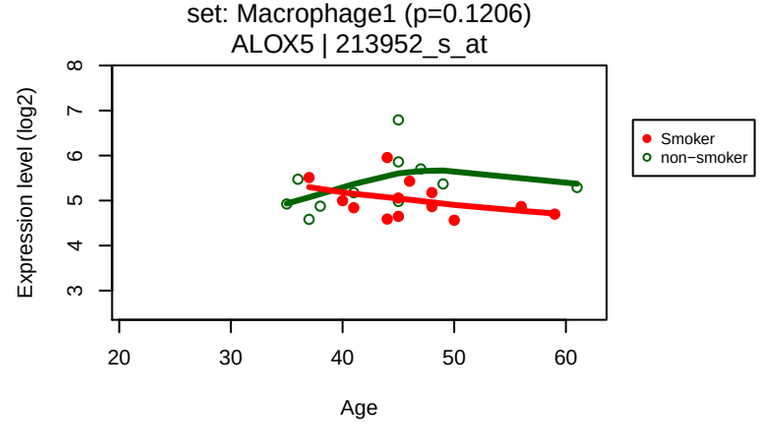

set: SmallAirway1 (p=0.2321)
ALOX5 | 213952_s_at
Expression level (log2)
Age
Smoker
non−smoker

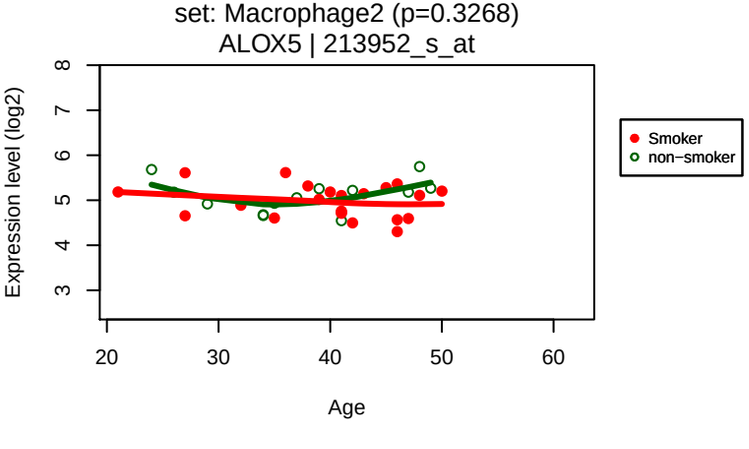

set: SmallAirway2 (p=0.1771)
ALOX5 | 213952_s_at
Expression level (log2)
Age
Smoker
non−smoker

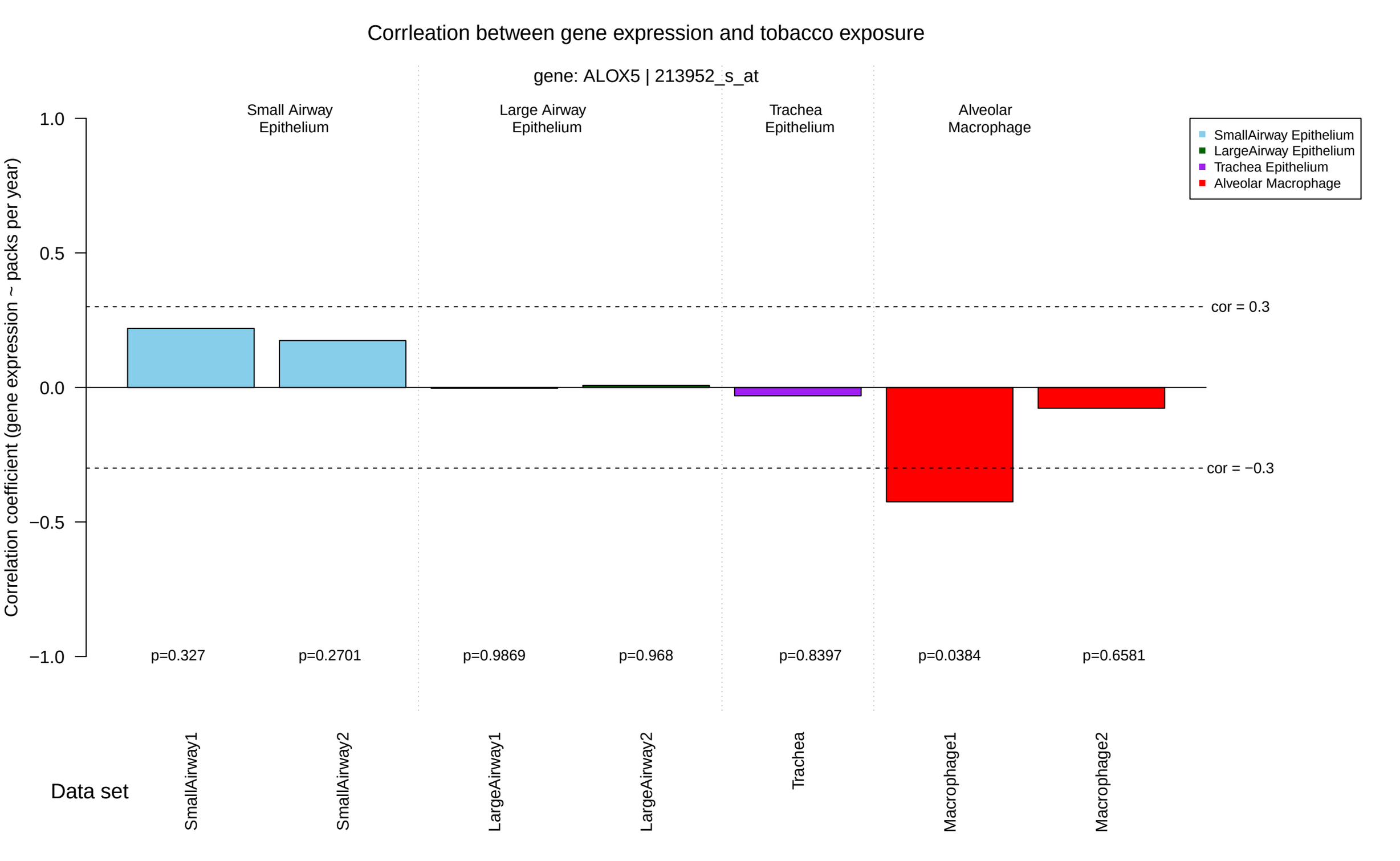

set: LargeAirway1 (p=0.047)
ALOX5 | 213952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1514)
ALOX5 | 213952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4033)
ALOX5 | 213952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1206)
ALOX5 | 213952_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3268)
ALOX5 | 213952_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOX5 | 213952_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.327
p=0.2701
p=0.9869
p=0.968
p=0.8397
p=0.0384
p=0.6581
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2